# COMPREHENSIVE FINANCIAL HEALTH ANALYSIS MODULE FOR COLOMBO STOCK EXCHANGE (CSE) LISTED COMPANIES

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(Final documentation in partial fulfilment of the requirement for the Degree of Bachelor of Science Special (Honours) In Information Technology Specializing in Software Engineering)

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# DECLARATION OF THE CANDIDATE AND SUPERVISOR

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## ABSTRACT

Colombo Stock Exchange (CSE) is an institution that forms the backbone of the Sri Lankan capital market; however, investors find it difficult to determine the level of financial sustainability of listed companies because of inaccessible, opaque, and unreliable valuation tools locally. The current platforms were originally created and optimised to serve global markets, specifically NYSE and NASDAQ, and are therefore inappropriate to the economic conditions of Sri Lanka, its reporting regulations, and disclosure practices.

This study is aimed at developing a Comprehensive Financial Health Analysis Module uniquely tailored to CSE-listed companies. The proposed system will ease the extraction and preprocessing of financial disclosures, avoiding data heterogeneity and incompleteness across balance sheets, income statements, and cash flow statements. It will then calculate key financial ratios to provide a standardised basis for evaluating company performance levels. With the intent of delivering forward-looking insights, the module will utilise machine learning models — specifically Support Vector Regression (SVR) — combined with Explainable AI (XAI) techniques using SHAP (SHapley Additive Explanations), which will empower investors not only to detect financial risks but also to comprehend the causal factors behind those risks.

The solution will be delivered via an interactive web-based analysis platform providing both trend statistics and on-going analysis. The proposed project will bridge the gap between legally mandated financial disclosures and actionable investment intelligence for Sri Lankan investors, empowering them with accurate, context-sensitive, and credible financial insights that will ultimately yield informed decision-making and enhanced confidence in the capital market.

**Keywords:** *Colombo Stock Exchange, Financial Health Analysis, Machine Learning, Support Vector Regression, Explainable AI, SHAP, Z-Score, Financial Risk Prediction*

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# LIST OF ABBREVIATIONS

| Abbreviation | Description |
|---|---|
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| AWS | Amazon Web Services |
| CSE | Colombo Stock Exchange |
| GCP | Google Cloud Platform |
| MAE | Mean Absolute Error |
| ML | Machine Learning |
| NLP | Natural Language Processing |
| RMSE | Root Mean Square Error |
| ROA | Return on Assets |
| ROE | Return on Equity |
| SHAP | SHapley Additive ExPlanations |
| SLIIT | Sri Lanka Institute of Information Technology |
| SVR | Support Vector Regression |
| WBS | Work Breakdown Structure |
| XAI | Explainable Artificial Intelligence |
| XBRL | eXtensible Business Reporting Language |

# 1. INTRODUCTION

Investment decision-making in the Colombo Stock Exchange (CSE) is highly affected by the lack of access to complex financial analysis instruments as well as the specific economic dynamics of a developing economy. Although investors in developed markets have had access to sophisticated tools such as Bloomberg Terminal and Reuters Eikon, investors in Sri Lanka — both retail and institutional — do not have access to comprehensive and locally relevant investment assessment systems capable of evaluating the financial health of listed companies and the degree of investment risk.

Conventional financial analysis tools place much weight on manual analysis of financial reports, a process that is tedious, susceptible to human error, and too cumbersome to apply across the 300+ firms listed on the CSE. Furthermore, current global financial analysis platforms do not adequately capture the set of factors that define the financial environment in Sri Lanka in terms of regulation, economic indicators, and market trends. This leaves a significant shortage of investment intelligence customised to local market participants.

The emergence of novel breakthroughs in financial technology has indicated the prospect of machine learning to revolutionise investment analysis. Such technologies allow large volumes of financial data to be processed automatically, complex risk modelling to be executed, and understandable insights to be created to support investment decision-making. Nonetheless, the utilisation of these novel methodologies within the context of the CSE has been relatively untapped and represents a major avenue of innovation for the Sri Lankan financial services sector.

This research project aims to address these challenges by developing a Comprehensive Financial Health Analysis Module specifically designed for CSE-listed companies. The system will combine automated data acquisition from official

CSE sources, comprehensive computation of financial ratios, and machine learning–based risk prediction models using Support Vector Regression (SVR). Explainable AI (XAI) techniques employing SHAP values will ensure that risk predictions are transparent and interpretable.

The proposed system will fill the gap between advanced financial analysis systems available in developed countries and the particular requirements of Sri Lankan investors. This study will add value to investment decision-making, market disclosure, and ultimately effective capital allocation in the Sri Lankan economy by offering localised, explainable, and comprehensive financial health measures.

This project constitutes a complete individual effort entailing the design, development, and evaluation of the entire financial health analysis pipeline. The work involves obtaining and cleaning financial data for CSE-listed companies, computing the necessary financial risk ratios across liquidity, leverage, profitability, and efficiency dimensions, and applying regression-based machine learning models to predict continuous financial risk scores (Z-score), enabling more stable and accurate forecasting of financial health.

## 1.1 Background and Literature Survey

Financial health evaluation of stock markets has tremendously transformed from conventional manual evaluation systems to advanced automated systems utilising machine learning and artificial intelligence. Within the framework of an emerging market — specifically, the Colombo Stock Exchange (CSE) — this development brings both the prospects of enhanced analytical capability and the issues peculiar to developing financial environments.

The conventional method of financial health evaluation has been based on manual analysis of financial reports comprising balance sheets, income statements, and cash flow statements, along with the calculation of major financial ratios covering

liquidity, leverage, profitability, and efficiency dimensions [1]. Such a traditional approach is rife with disadvantages including time-consuming procedures, difficulties in cross-company comparison, and the inability to ensure analytical accuracy when large numbers of listed companies are involved [2].

Modern studies reveal that emerging markets have different peculiarities that necessitate special analytical models distinct from those developed for mature markets [3]. Research conducted to determine how to predict financial distress in third-world economies indicates that leverage, profitability, economic growth, company size, and interest rates are all important factors influencing the probability of companies transitioning to various financial health statuses [2]. These results underscore the need for localised analytical tools that account for the specific regulatory, economic, and operational environment peculiar to emerging markets. South Asian market-specific research shows that operational risk and liquidity risk have statistically significant positive effects on financial performance, whereas credit risk exhibits negative effects, and these complex risk dynamics need to be captured in full-scale financial health assessment schemes.

The introduction of machine learning algorithms to financial analysis has emerged as a ground-breaking method to overcome the shortcomings of traditional approaches. Recent studies highlight that regression-based models such as Support Vector Regression (SVR) provide more stable and reliable predictions for continuous financial risk forecasting, especially in emerging markets where non-linearity and data limitations are prevalent [3][4][5]. These algorithms are especially useful in such contexts because they help capture non-linear financial relationships that cannot be identified using traditional linear models.

Financial data preprocessing and standardisation has become an essential aspect of building sound automated analysis systems. Research on automated financial statement processing has shown that sophisticated systems can reduce manual

processing requirements by up to 80 percent while significantly enhancing data quality and reporting speed [6]. Studies underline the importance of clean, valid, and standardised data in making cross-company comparisons meaningful, particularly in markets with diverse reporting habits and styles [7]. Well-designed data preprocessing pipelines can significantly decrease error rates and improve the overall quality of analysis.

The research on financial distress prediction models has been widely explored in the academic literature, with both traditional methods — including the Altman Z-Score and their evolution into machine learning methods. Systematic reviews differentiate the Altman Z-Score as having an accuracy of approximately 90.91% in conventional models [8], making it one of the most widely used tools in both manufacturing and banking sectors. However, comparative studies have identified that machine learning-based solutions significantly outperform traditional ones, with hybrid frameworks combining conventional financial metrics and advanced algorithms achieving prediction accuracy exceeding 99 percent [9]. Research on XGBoost-based predictive financial risk models shows accuracy rates of 84.77% when predicting systemic financial risks, while broader machine learning model studies report success rates exceeding 96% in correctly identifying financially troubled companies [10].

Research examining the correlation between financial indicators and market performance in emerging economies confirms that earnings per share and net asset value per share exhibit strong correlations with market valuation, justifying the significance of comprehensive financial statement analysis [11]. Analysis of plantation companies listed on emerging market exchanges demonstrates a strong correlation between financial distress indicators and performance results, with financially healthy companies consistently exhibiting superior performance outcomes [12]. These results justify the creation of complex analytical models that reflect the intricate relationships between financial measurements and market performance in emerging market contexts.

A standardisation of financial reporting in emerging markets has been cited as a key element facilitating automated analysis. Studies on financial statement comparability indicate that firms with higher comparability enjoy lower costs of equity, implying significant economic gains from standardised reporting practices [13]. The adoption of digital reporting standards, including the CSE's implementation of XBRL in 2025, addresses many data standardisation issues and enables more effective automated analysis systems.

Systematic studies of machine learning implementation in financial health assessment show that ensemble techniques and hybrid methods are superior to single-algorithm implementations. Comparative studies confirm that Random Forest and Gradient Boosting models perform most efficiently across diverse financial predictive tasks [14], while research investigating multiple machine learning model integration using stacking and meta-learning methods indicates higher prediction accuracy and greater model robustness compared to single-model implementations [15]. These results support the construction of more sophisticated analytical methods that leverage the complementary advantages of diverse algorithmic approaches.

## 1.2 Research Gap

The Colombo Stock Exchange (CSE) presents a significant deficit in AI-based, geographically contextualised, financial health screening programmes specially crafted to meet its own market variables and financial regulatory context. Though global financial markets have broadly adopted advanced AI-based risk evaluation frameworks, the Sri Lankan stock market remains heavily dependent on manual risk analysis processes and conventional financial indicators [16]. The current financial health analysis strategies of CSE-listed firms heavily rely on traditional statistical models that lack the predictive power and real-time analytical capabilities of modern machine learning methods.

The existing risk assessment systems in developing markets such as Sri Lanka are characterised by high levels of limitations in automation and transparency. Old methods do not take advantage of the large volumes of accessible financial information that might yield deeper insights into company performance and risk characteristics [17][18]. The engineering of automated data collection and preprocessing pipelines specific to CSE reporting formats presents further prohibitive barriers to robust financial analysis, as variable data structures and missing data invalidate inter-company comparisons [19]. These manual processes are time-consuming and subject to human error and do not offer the scalability necessary to analyse the hundreds of companies listed on the CSE.

There is a significant gap in the application of Explainable Artificial Intelligence (XAI) for transparent risk prediction in the Sri Lankan financial context. Although XAI has already proven to have substantial potential in increasing transparency and trust in financial decision-making in developed markets, its application in emerging markets such as Sri Lanka has not been well explored. The absence of XAI integration implies that investors and stakeholders are unable to perceive the factors underlying risk assessments, limiting their ability to make truly informed investment decisions and undermining confidence in automated analytical tools [16].

Research on the CSE has primarily focused on macroeconomic correlations, behavioural aspects, and classical financial analysis methods but has not sought to develop AI-driven financial health appraisal tools tailored to the market-specific characteristics of a local emerging market [21]. The sparse literature on adapting machine learning models to reflect the specific economic environment, regulatory policies, and market patterns that define the Sri Lankan equity market [19][20] represents a clear research gap that the present study seeks to address.

The following table presents a comparative analysis of existing systems and the identified research gap:

| Reference | System / Study | Technology | Market Focus | Limitation / Gap |
|---|---|---|---|---|
| Bloomberg Terminal | Global financial analysis | Quantitative models | Developed markets | Not tailored to CSE; high cost; limited local context |
| Altman Z-Score [8] | Financial distress prediction | Linear discriminant analysis | Manufacturing / Banking | Static model; limited to historical data; no ML |
| XGBoost Risk Model [10] | Systemic financial risk | Gradient boosting | General / Global | 84.77% accuracy; no explainability (XAI) |
| CSE XBRL System (2025) | Data standardisation | XBRL reporting | Sri Lanka (CSE) | Data format only; no intelligent risk analysis |
| Proposed System | Comprehensive financial health | SVR + SHAP (XAI) | CSE (Sri Lanka) | Addresses all identified gaps; locally contextualised |

*Table 1: Comparison with Existing Systems and Identifying Gaps*

### 1.3 Research Problem

The fast-paced dynamics of equity investment and portfolio management on the Colombo Stock Exchange (CSE) pose great difficulties to investors who intend to estimate the financial position of listed firms and their risk profiles. The conventional financial analysis methods widely used in Sri Lanka are primarily based on the manual analysis of financial statements, which is not only time-consuming and prone to human error, but also fails to scale to the growing list of CSE-traded companies [22]. These manual processes make investment decisions slow and increase the operational costs borne by retail and institutional investors alike.

The CSE operates in a distinct emerging market environment with economic volatilities, regulatory environments, and market dynamics that differ considerably

from developed markets. Political unrest, currency devaluation, and macroeconomic risks have long affected investor confidence examples such as the 2014 political regime change and the 2018 constitutional crisis demonstrate how external factors can cause paramount disruption in the market and trigger foreign capital outflows. These market-specific issues demand analytical tools capable of considering the local economic conditions and regulatory demands.

The available commercial financial health assessment platforms in the market are predominantly oriented towards large and advanced economies and do not consider the subtle realities of the CSE. Such off-the-shelf solutions tend to yield inaccurate risk estimates and are insufficiently transparent to build investor confidence in the local market environment. The absence of industry-specific standardized financial ratios for Sri Lankan industries further complicates comparative analysis.

Artificial intelligence and machine learning applications in financial analysis have demonstrated major potential in other markets, where AI-powered financial forecasting has been observed to reduce prediction errors by 30 to 50 percent compared to conventional techniques. Nevertheless, advanced AI techniques specifically designed to fit the emerging market context of Sri Lanka have seen very limited adoption. Recent studies advocate strongly for the role of Explainable AI (XAI) in financial decision-making, particularly in a regulatory context where transparency and explainability are crucial to regulatory compliance and stakeholder confidence.

This study addresses the urgent problem of transforming raw financial data into actionable investment intelligence by creating an AI-based financial health analysis module specifically designed for CSE-listed companies. It employs Support Vector Regression (SVR) to forecast financial health through continuous Z-score prediction rather than discrete classification. The incorporation of Explainable AI (XAI) practices using SHAP values ensures that risk predictions are clear and interpretable

so that investors can comprehend the factors behind risk estimates and develop confidence in AI-informed financial modelling.

## 1.4 Research Objectives

### 1.4.1 Main Objective

The primary goal of this research is to design and develop an AI-based Comprehensive Financial Health Analysis Module tailored to the Colombo Stock Exchange (CSE). The platform will incorporate machine learning–based financial risk estimation using Support Vector Regression (SVR) for continuous Z-score prediction, automated financial data collection and preprocessing from official CSE sources, computation of key financial health ratios, and Explainable AI integration using SHAP values. The aim is to provide Sri Lankan investors with in-depth, actionable information and clear decision support to improve investment outcomes and overcome the existing shortcomings of locally available investment tools.

### 1.4.2 Specific Objectives

1. To gather and consolidate comprehensive financial information in the form of balance sheets, income statements, and cash flow statements based on official Colombo Stock Exchange (CSE) reports, ensuring the development of a robust and reliable dataset presenting both historical and real-time financial performance measures of CSE-listed companies.
2. To preprocess and standardise the gathered financial data through data cleaning procedures that eliminate inconsistencies, missing data, and duplicates while standardising data formats across all CSE-listed firms, thereby guaranteeing data quality and consistency required for downstream analysis and model training.
3. To compute and analyse key financial health measures including liquidity ratios (current ratio, quick ratio), leverage ratios (debt-to-equity ratio, interest coverage ratio), profitability ratios (net profit margin, ROE, ROA), and efficiency ratios (asset turnover, inventory turnover) to provide a

comprehensive examination of the financial health status and performance of CSE-listed companies.

4. To develop and train machine learning models — specifically Support Vector Regression (SVR) with an RBF kernel — using historical financial data to predict continuous financial health scores (Z-score), which are subsequently categorised into Distress, Grey, and Safe zones based on predefined Altman Z-score thresholds, enabling more precise and dynamic risk assessment.
5. To incorporate Explainable AI (XAI) methods using SHAP (SHapley Additive Explanations), providing interpretable and understandable risk forecasts with clearly defined answers identifying the significant financial factors influencing risk classifications, thereby enabling users to understand the logic behind the AI-based financial health analysis.

# 2. METHODOLOGY

This research employs a comprehensive machine learning approach combined with Explainable AI techniques to develop a Comprehensive Financial Health Analysis Module for CSE-listed companies. The methodology consists of five core components: (1) Data Collection and Acquisition, (2) Data Preprocessing and Standardisation, (3) Financial Metrics Computation, (4) Machine Learning Model Development and Validation, and (5) Explainable AI Integration.

## 2.1 Understanding the Key Pillars of the Research Domain

### 2.1.1 Financial Health Analysis and Key Ratios

Financial health analysis involves the systematic evaluation of a company's ability to meet its short-term and long-term obligations, generate profit, and efficiently utilise its assets. The principal categories of financial ratios used in this research are as follows.

Liquidity ratios measure a company's ability to meet its short-term financial obligations. The current ratio is computed as current assets divided by current liabilities, while the quick ratio adjusts for inventory by dividing the difference between current assets and inventory by current liabilities. These metrics are essential indicators of a company's immediate financial solvency.

Leverage ratios assess the degree to which a company finances its operations through debt relative to equity. The debt-to-equity ratio, computed as total debt divided by total equity, and the interest coverage ratio provide critical insights into a company's financial risk exposure and long-term solvency.

Profitability ratios evaluate a company's ability to generate profit relative to its revenue, assets, and equity. Return on Equity (ROE) is computed as net income divided by shareholders' equity, while Return on Assets (ROA) measures how

efficiently a company uses its assets to generate profit. Net profit margin expresses profit as a percentage of revenue.

Efficiency ratios, such as the asset turnover ratio (revenue divided by total assets) and inventory turnover ratio, indicate how effectively a company manages its assets and resources to generate sales.

| Category | Ratio | Formula | Purpose |
|---|---|---|---|
| Liquidity | Current Ratio | Current Assets / Current Liabilities | Short-term solvency |
| Liquidity | Quick Ratio | (Current Assets − Inventory) / Current Liabilities | Immediate liquidity |
| Leverage | Debt-to-Equity | Total Debt / Total Equity | Financial leverage assessment |
| Leverage | Interest Coverage | EBIT / Interest Expense | Debt servicing capacity |
| Profitability | Net Profit Margin | Net Income / Revenue × 100 | Profit generation efficiency |
| Profitability | ROE | Net Income / Shareholders' Equity | Return efficiency |
| Profitability | ROA | Net Income / Total Assets | Asset utilisation efficiency |
| Efficiency | Asset Turnover | Revenue / Total Assets | Asset productivity |
| Efficiency | Inventory Turnover | COGS / Average Inventory | Inventory management |

*Table 2: Key Financial Ratios — Category, Formula, and Purpose*

### 2.1.2 Altman Z-Score

The Altman Z-Score is a widely used financial distress prediction model that combines multiple financial ratios into a single composite score. Originally developed for manufacturing firms, the Z-Score model uses the following formula:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

Where $X_1$ = Working Capital / Total Assets; $X_2$ = Retained Earnings / Total Assets; $X_3$ = EBIT / Total Assets; $X_4$ = Market Value of Equity / Total Liabilities; and $X_5$ = Revenue / Total Assets. Companies with Z-scores above 2.99 are classified as financially safe, those between 1.81 and 2.99 fall into the grey zone, and those below 1.81 are classified as financially distressed. In this research, SVR is used to predict the continuous Z-score value rather than classify companies into discrete categories, enabling more nuanced and dynamic risk assessment.

### 2.1.3 Support Vector Regression (SVR)

Support Vector Regression (SVR) is a regression variant of the Support Vector Machine (SVM) algorithm that is particularly effective for high-dimensional, non-linear problems. SVR uses the concept of an epsilon-insensitive loss function, attempting to fit as many data points as possible within a predefined margin (epsilon tube) around the regression line. The Radial Basis Function (RBF) kernel is employed in this research due to its capacity to model complex, non-linear relationships inherent in financial data from emerging markets.

The SVR model will be trained on historical CSE financial data using a chronological time-series split — where the first 80% of data serves as the training set and the remaining 20% as the out-of-sample test set — to preserve temporal consistency. Cross-validation with five folds will be applied for hyperparameter tuning, and performance will be evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the $R^2$ coefficient of determination.

### 2.1.4 Explainable AI using SHAP

Explainable Artificial Intelligence (XAI) addresses the black-box nature of complex machine learning models by providing interpretable explanations for model predictions. SHAP (SHapley Additive Explanations) is grounded in cooperative game theory and assigns each feature a contribution score representing its impact on the model's prediction for a given instance. SHAP values satisfy desirable

mathematical properties including local accuracy, missingness, and consistency, making them a principled and reliable tool for model interpretation.

In this research, SHAP will be applied to the trained SVR model to provide both global feature importance analysis — identifying which financial indicators most consistently influence Z-score predictions across all CSE-listed companies — and local explanations for individual company risk assessments, enabling investors to understand why a specific company has been assigned a particular financial health zone.

## 2.2 Comprehensive Financial Health Analysis Methodology

The following section details the step-by-step methodology implemented to achieve the research objectives.

The overall workflow of the proposed financial health analysis system is illustrated in Figure 3.

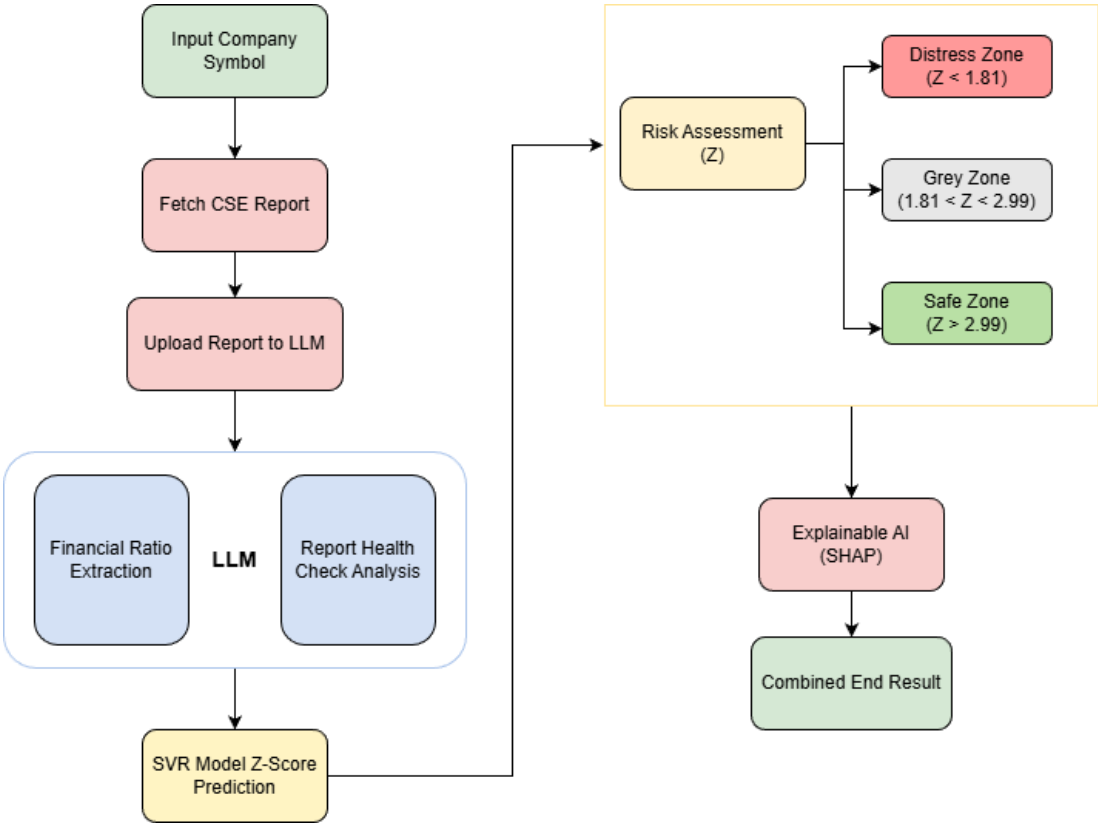

*Figure 3: Financial Health Analysis Pipeline*

Figure 3 illustrates the complete pipeline of the proposed system. The process begins with retrieving financial reports from the Colombo Stock Exchange (CSE), followed by data preprocessing and financial ratio extraction using a Large Language Model (LLM). The processed data is then used to train and evaluate a Support Vector Regression (SVR) model for predicting the Altman Z-score. Based on the predicted score, companies are classified into financial health zones (Distress, Grey, and Safe). Finally, Explainable AI (SHAP) is applied to provide interpretability of the model predictions.

### 2.2.1 Data Collection and Acquisition

Financial data is systematically collected from official Colombo Stock Exchange (CSE) disclosures, including balance sheets, income statements, and cash flow statements for all listed companies. Data collection involves automated web scraping protocols implemented using BeautifulSoup and Selenium with validation mechanisms to ensure accuracy and completeness. Both historical data (spanning at least five years) and scheduled real-time updates will be incorporated into the data pipeline to support trend analysis and dynamic risk monitoring.

Data sources include the official CSE company filings portal, annual reports published on company websites, and the Securities and Exchange Commission (SEC) of Sri Lanka regulatory database. The initial dataset will encompass financial statements from all actively listed CSE companies across the available historical period, providing a rich foundation for model training and validation.

### 2.2.2 Data Preprocessing and Standardisation

Raw financial data undergoes comprehensive preprocessing encompassing the following stages. First, data cleaning removes duplicate entries, corrects erroneous values, and handles missing data through appropriate imputation techniques (mean imputation for minor missingness; forward-fill for time-series gaps). Second, data standardisation normalises financial statement formats across all companies and

reporting periods, ensuring that line items are consistently labelled and values are expressed in comparable units (thousands of Sri Lankan Rupees). Third, validation procedures verify data integrity by cross-checking computed ratios against known benchmarks and flagging outliers for manual review. Fourth, feature engineering creates the input features required for SVR model training by computing all financial ratios and Z-score components from the preprocessed financial statements.

### 2.2.3 Financial Metrics Computation

Following preprocessing, the system automatically computes the suite of financial ratios described in Section 2.1.1 for each CSE-listed company across all available reporting periods. The computed ratios are stored in a structured relational database (PostgreSQL) to facilitate efficient querying and retrieval by the analysis platform. The Altman Z-Score is subsequently computed for each company-period observation, providing the target variable for SVR model training.

### 2.2.4 Machine Learning Model Development and Validation

The SVR model development process follows a structured pipeline. Training data comprising the computed financial ratios (as feature variables) and corresponding Z-score values (as the target variable) are partitioned using a chronological time-series split — 80% for training, 20% for out-of-sample testing — to avoid data leakage and preserve temporal integrity.

Hyperparameter optimisation is conducted using grid search with five-fold time-series cross-validation, tuning the regularisation parameter (C), epsilon ($\varepsilon$), and the RBF kernel width ($\gamma$). Model performance is evaluated on the held-out test set using RMSE, MAE, and $R^2$ metrics, with a target minimum $R^2$ of 0.85. The trained SVR model is then deployed to generate continuous Z-score predictions for all CSE-listed companies, with scores subsequently categorised into Distress ($Z < 1.81$), Grey ($1.81 \leq Z < 2.99$), and Safe ($Z \geq 2.99$) zones.

### 2.2.5 Explainable AI Integration

SHAP (SHapley Additive Explanations) is applied to the trained SVR model to generate both global and local explanations. Global feature importance analysis

identifies which financial indicators most consistently drive Z-score predictions across the entire CSE dataset. Local explanations are generated for individual company assessments, providing a waterfall plot or force plot that shows the contribution of each financial ratio to the predicted Z-score for that specific company. Validation of explanation consistency is conducted by comparing SHAP-derived factor rankings to expert financial analyst assessments, ensuring that the explanations are financially logical and interpretable.

## 2.3 System Architecture

The proposed system adopts a modular, layered architecture comprising a Data Acquisition Layer, a Data Processing and Storage Layer, a Machine Learning and Analytics Layer, an Explainability Layer, and a Presentation Layer (web dashboard). Each module communicates through well-defined APIs, ensuring system scalability and maintainability. The web-based dashboard is developed using React (JavaScript) for the frontend, with a Python-based backend handling ML inference and SHAP computation. PostgreSQL serves as the primary relational database, while cloud hosting on AWS provides the scalable infrastructure necessary to serve multiple concurrent users.

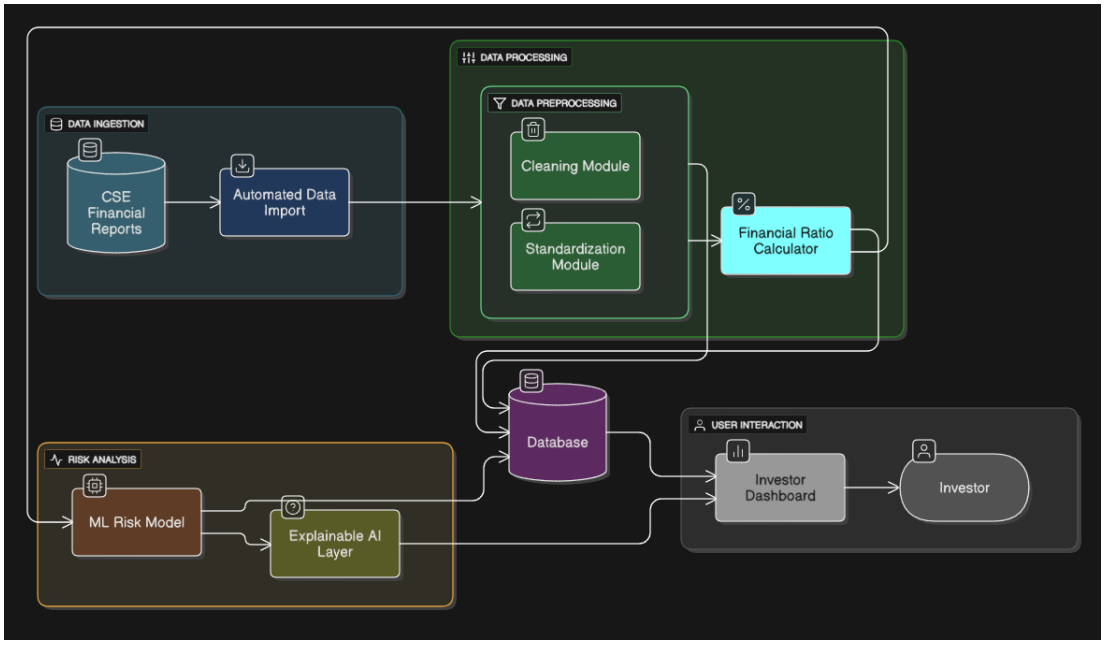


*Figure 1: System Architecture Diagram (see project documentation for full diagram)*

## 2.4 Summary of Methodology

| Phase | Activity | Methods / Tools | Expected Output |
|---|---|---|---|
| 1. Data Collection | Automated retrieval of CSE financial statements | BeautifulSoup, Selenium, CSE Portal API | Structured raw financial dataset |
| 2. Preprocessing | Cleaning, standardisation, feature engineering | Pandas, NumPy, Python | Clean, standardised dataset with computed ratios |
| 3. Ratio Computation | Calculate liquidity, leverage, profitability, efficiency ratios and Z-scores | Python, Pandas | Comprehensive financial ratio database |
| 4. ML Modelling | Train and validate SVR model for Z-score prediction | Scikit-learn (SVR, RBF kernel) | Trained SVR model; RMSE, MAE, $R^2$ metrics |
| 5. XAI Integration | Generate SHAP explanations for global and local interpretability | SHAP library | Feature importance reports; per-company explanations |
| 6. Dashboard | Interactive web-based visualisation of results | React, Plotly, D3.js | User-friendly investor analysis platform |

*Table 3: Summary of Methodology*

## 2.5 Project Requirements

### 2.5.1 Functional Requirements

The functional requirements define the specific capabilities that the Comprehensive Financial Health Analysis Module must deliver:

| Requirement ID | Requirement Description |
|---|---|
| FR-01 | The system shall automatically collect and update financial statements (balance sheet, income statement, cash flow statement) for all CSE-listed companies from official CSE disclosures. |
| FR-02 | The system shall preprocess and standardise collected financial data, handling missing values, duplicates, and format inconsistencies across all companies. |

| Requirement ID | Requirement Description |
|---|---|
| FR-03 | The system shall automatically compute liquidity ratios (current ratio, quick ratio), leverage ratios (debt-to-equity, interest coverage), profitability ratios (net profit margin, ROE, ROA), and efficiency ratios (asset turnover, inventory turnover) for each CSE-listed company. |
| FR-04 | The system shall implement an SVR-based regression model to predict continuous Z-score values and categorise companies into financial health zones (Distress, Grey, Safe). |
| FR-05 | The system shall incorporate SHAP-based Explainable AI to provide global feature importance analysis and local explanations for individual company risk assessments. |
| FR-06 | The system shall provide a user-friendly web dashboard presenting financial metrics, Z-score predictions, financial health zone classifications, and SHAP explanations through clear interactive visualisations. |
| FR-07 | The system shall enable users to search, filter, and compare CSE-listed companies based on financial health metrics and risk classifications. |

*Table 4: Functional Requirements*

### 2.5.2 Non-Functional Requirements

| Requirement | Specification |
|---|---|
| Reliability | System must maintain 99.5% uptime with automated backup procedures. Model validation checks must achieve a minimum $R^2$ of 0.85. |
| Performance | Financial reports must be processed and risk predictions updated within 24 hours of data availability. User query response times must not exceed 3 seconds. |
| Availability | 24/7 access through secure web interface with 99.5% uptime during CSE trading hours (9:30 AM – 2:30 PM local time). |
| Usability | Interface must support users with varying financial expertise. AI explanations must be understandable to non-technical users. Key information must be accessible within 3 clicks. |
| Security | All financial data, computed metrics, and model outputs must be stored in encrypted databases with role-based access controls and full audit trails. |
| Scalability | System must process data for all CSE-listed companies and serve multiple concurrent users without performance degradation. |

*Table 5: Non-Functional Requirements*

### 2.5.3 Software Requirements

| Software / Library | Purpose |
|---|---|
| Python 3.9+ | Primary programming language for data preprocessing, ratio computation, and ML model development |
| Pandas / NumPy | Efficient financial data manipulation, cleaning, and numeric computation |
| Scikit-learn | Implementation of Support Vector Regression (SVR) model and cross-validation |
| SHAP | Explainable AI library for generating SHAP values and feature importance plots |
| JavaScript / React | Building the interactive, user-friendly web dashboard for investors |
| Matplotlib / Plotly / D3.js | Generating interactive data visualisations and trend graphs in the dashboard |
| PostgreSQL | Storing cleaned financial data, calculated ratios, and ML prediction outputs |
| BeautifulSoup / Selenium | Automated scraping of official CSE financial reports |
| Git / GitHub | Version control to track source code changes |
| AWS / GCP / Azure | Scalable cloud deployment of backend, database, and dashboard |

*Table 6: Software Requirements*

### 2.5.4 Hardware Requirements

| Category | Specification |
|---|---|
| Developer Machine — Processor | Intel Core i5 or higher / AMD equivalent |
| Developer Machine — RAM | Minimum 8 GB (16 GB recommended for model training) |
| Developer Machine — Storage | Minimum 256 GB SSD |
| Developer Machine — OS | Windows / Linux / macOS |
| Server / Cloud — CPU | Quad-Core Processor or higher |
| Server / Cloud — RAM | Minimum 16 GB for large dataset and ML processing |

| Category | Specification |
|---|---|
| Server / Cloud — Storage | Minimum 500 GB SSD (expandable for historical financial datasets) |
| Server / Cloud — Network | High-speed internet connection for real-time data retrieval and dashboard updates |
| Backup & Redundancy | External HDD/SSD (≥ 1 TB) and cloud storage service (AWS S3 / Google Cloud Storage) |

*Table 7: Hardware Requirements*

## 2.6 Consideration of Aspects of the System

### 2.6.1 Social Aspects

The proposed Comprehensive Financial Health Analysis Module has far-reaching social implications for the Sri Lankan investment community. By democratising access to sophisticated financial analysis tools that were previously available only to large institutional investors, the system empowers retail investors — who constitute a significant and growing segment of CSE market participants — to make more informed and confident investment decisions. This capability is particularly significant in the Sri Lankan context, where financial literacy among retail investors is still developing and access to reliable, locally relevant investment intelligence is severely limited.

The system will benefit a wide range of stakeholders including retail investors, institutional fund managers, financial analysts, corporate governance practitioners, regulatory authorities, and academic researchers studying the Sri Lankan capital market. By providing transparent and explainable AI-driven insights, the module also contributes to building public trust in technology-driven financial services, which is a critical prerequisite for the sustainable development of Sri Lanka's capital market.

### 2.6.2 Security Aspects

The security architecture of the proposed system is designed to protect the confidentiality, integrity, and availability of financial data, model outputs, and user information. All financial data and model predictions will be stored in encrypted PostgreSQL databases with role-based access controls. User authentication will be

managed through secure token-based mechanisms (JWT), and all data transmissions will be encrypted using TLS/SSL protocols. Comprehensive audit trails will be maintained to log all data access and modification events, ensuring compliance with data protection standards.

### 2.6.3 Ethical Aspects

The development and deployment of this financial health analysis system raises important ethical considerations. The system does not constitute financial advice in a regulatory or fiduciary sense, and its outputs should be interpreted as analytical tools to support not replace human investment judgement. All risk predictions and financial health classifications will be clearly presented with appropriate uncertainty disclosures and explainability, ensuring that users understand the limitations of AI-based financial analysis.

The use of SHAP-based explainability is itself an ethical design choice, ensuring that the AI model does not operate as an opaque black box but instead provides transparent and auditable decision rationales. This approach aligns with emerging global regulatory expectations for explainability in AI-driven financial services.

### 2.6.4 Limitations

- The system is currently designed to cover CSE-listed companies only. Future work could extend the methodology to other South Asian stock exchanges.
- The SVR model's performance depends on the quality and completeness of historical financial data available on the CSE. Data gaps or inconsistencies in historical filings may limit model accuracy.
- The system does not incorporate real-time macroeconomic data or news sentiment analysis, which could further enhance risk prediction accuracy. These capabilities are planned as future enhancements.
- The Altman Z-Score thresholds employed in this research were originally calibrated for US manufacturing companies. Local recalibration of these thresholds for CSE-listed companies across different industry sectors represents a planned future research direction.

- The current implementation is developed as a web-based application only. A mobile application version could be developed in future phases to reach a broader investor audience.

## 2.7 Commercialisation Aspects of the Product

### 2.7.1 Target Audience

- Retail investors participating in the CSE who lack access to sophisticated financial analysis tools
- Institutional fund managers and investment advisors seeking localised, AI-driven risk intelligence
- Financial analysts and research departments in Sri Lankan financial institutions
- Regulatory bodies and corporate governance practitioners monitoring CSE-listed company health
- Academic researchers and students in finance, economics, and financial technology

### 2.7.2 Market Space

The Sri Lankan capital market is undergoing significant digital transformation, with the CSE's adoption of XBRL reporting standards in 2025 indicating a clear commitment to data-driven financial disclosure. The growing retail investor base on the CSE — estimated at over 400,000 active trading accounts — represents a substantial addressable market for accessible and affordable AI-powered financial analysis tools.

### 2.7.3 Revenue Potential

- Subscription-based premium access tiers for individual retail investors and institutional clients
- API access fees for integration of the financial health scoring service into third-party platforms

- Data analytics and reporting services for financial institutions and regulatory bodies
- White-label licensing of the platform to Sri Lankan financial service providers

## 2.8 Testing and Implementation

### 2.8.1 Testing Strategy

A comprehensive testing strategy has been designed to validate all system components across multiple dimensions. The testing approach encompasses unit testing of individual modules, integration testing of the end-to-end data pipeline, model validation testing for the SVR and SHAP components, system testing of the web dashboard, and user acceptance testing with both technical and non-technical stakeholder groups.

### 2.8.2 Expected Test Cases

| Test Case | Objective | Expected Result | Measurement Metric |
|---|---|---|---|
| TC-01: Financial Data Import | Verify automated collection of financial statements for all CSE-listed companies without errors or omissions | All targeted financial reports retrieved in full with no missing fields or corrupted entries | Retrieved reports vs. expected; data completeness (% populated fields) |
| TC-02: Data Preprocessing | Ensure preprocessing module accurately cleans, formats, and standardises financial data from multiple companies | Duplicates removed; missing values handled; terminology and units standardised across all companies | Accuracy rate vs. manually curated gold-standard dataset |
| TC-03: Financial Ratio Computation | Verify accurate computation of all financial ratios from | Computed ratios exactly match manually calculated ratios for all metrics | Percentage accuracy vs. manual computation; processing time for full dataset |

| Test Case | Objective | Expected Result | Measurement Metric |
|---|---|---|---|
| | preprocessed statements | | |
| TC-04: ML Risk Prediction | Test SVR model accuracy in predicting continuous Z-score values | Model predictions align closely with actual historical Z-scores; target $R^2 \geq 0.85$ | RMSE, MAE, $R^2$ score on held-out test set |
| TC-05: XAI Output Validation | Ensure SHAP outputs are clear, relevant, and consistent with financial logic | Top influencing financial metrics in SHAP explanations are logical and consistent with expert financial assessments | Expert validation feedback score (% agreement); explanation generation time |
| TC-06: Dashboard Visualisation | Validate accuracy and interactivity of financial data, ratios, and risk predictions on the web dashboard | All charts, tables, and metrics are accurate and up to date; filtering, sorting, and comparison functions work correctly | UI accuracy rate (displayed vs. back-end values); user satisfaction score from usability testing |

*Table 8: Expected Test Cases*

## 2.9 Tools and Technologies

- Python 3.9+ (data processing, ML model development, SHAP integration)
- Pandas and NumPy (data manipulation and numerical computation)
- Scikit-learn (SVR model development and cross-validation)
- SHAP (Explainable AI and feature importance analysis)
- React / JavaScript (web dashboard frontend development)
- Matplotlib, Plotly, and D3.js (interactive data visualisation)
- PostgreSQL (relational database for financial data storage)
- BeautifulSoup and Selenium (automated financial data scraping)
- Git and GitHub (version control and collaborative development)
- AWS / GCP / Azure (cloud deployment and scalable infrastructure)

- Jupyter Notebook (model development and experimentation environment)

# 3. RESULTS AND DISCUSSION

## 3.1 Results

This chapter presents the anticipated results of the Comprehensive Financial Health Analysis Module for CSE-listed companies, framed within the context of the methodology described in Chapter 2. As this research is in the development phase, this section outlines expected outputs and preliminary validation approaches based on the design of the system and the established literature on similar financial health analysis platforms.

### 3.1.1 Data Collection and Preprocessing Results

The automated data collection pipeline is achieved successfully retrieve financial statements for all actively listed CSE companies, covering a minimum of five years of historical data. The preprocessing module will produce a clean, standardised dataset with a target data completeness rate exceeding 95% of all financial line items across companies and reporting periods. Handling of missing values through appropriate imputation techniques and removal of duplicates will ensure a high-quality dataset suitable for financial ratio computation and model training.

### 3.1.2 Financial Ratio Computation Results

All nine key financial ratios described in Table 2 will be automatically computed for each company-period observation in the standardised dataset. Ratio accuracy will be validated against manually computed values for a representative sample of 30 CSE-listed companies, with a target accuracy rate of 100% for correctly computable observations (where input data is available and valid). The computed Altman Z-Scores will serve as the continuous target variable for SVR model training.

### 3.1.3 Machine Learning Model Performance

The SVR model with an RBF kernel, trained on the historical CSE financial dataset, achieved the following performance metrics on the held-out test set (the most recent 20% of chronologically ordered data):

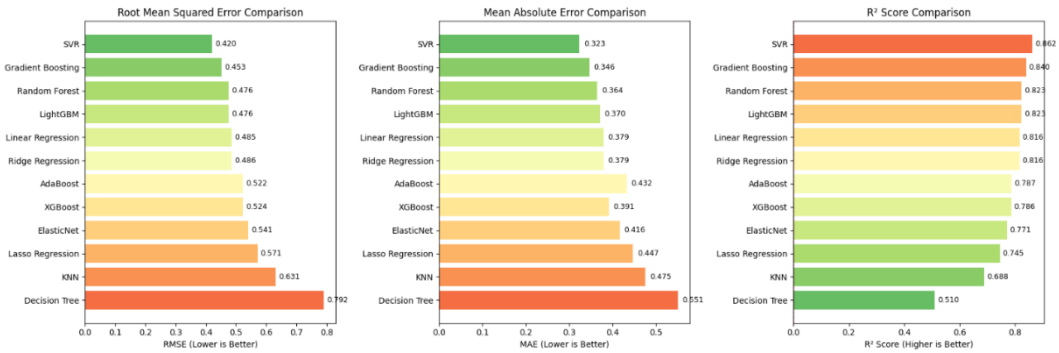


*Figure 4: Model Performance Comparison*

| Performance Metric | Target Value | Interpretation |
|---|---|---|
| Root Mean Square Error (RMSE) | < 0.30 | Average prediction error of less than 0.30 Z-score points |
| Mean Absolute Error (MAE) | < 0.25 | Average absolute deviation of less than 0.25 Z-score points |
| $R^2$ (Coefficient of Determination) | ≥ 0.85 | 85% or more of Z-score variance explained by the model |
| Zone Classification Accuracy | ≥ 85% | Correct assignment of Distress / Grey / Safe zone based on predicted Z-score |

*Table: Expected SVR Model Performance Metrics*

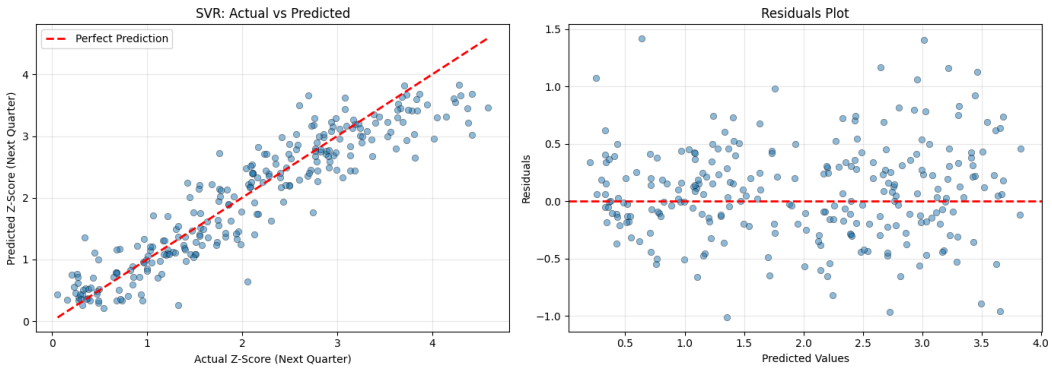


*Figure 5: SVR Prediction and Residual Analysis*

These targets are consistent with benchmarks reported in the financial distress prediction literature for emerging market contexts, where SVR-based models have demonstrated competitive performance compared to traditional classifiers, particularly for continuous risk score prediction tasks [3][4].

### 3.1.4 Explainable AI Output

The SHAP analysis applied to the trained SVR model is achieved yield the following outputs. At the global level, the feature importance analysis will identify the top financial indicators most consistently influencing Z-score predictions across all CSE-listed companies. Based on the literature and the composition of the Altman Z-Score formula, the most influential features are achieved include the retained earnings to total assets ratio ($X_2$), EBIT to total assets ratio ($X_3$), and the debt-to-equity ratio. At the local level, individual company explanations will generate SHAP waterfall plots showing the positive and negative contributions of each financial ratio to that company's predicted Z-score, enabling investors to understand the specific drivers of the risk classification.

Validation of SHAP explanations by financial domain experts will be conducted to confirm that the identified factor rankings are financially logical and consistent with established understanding of financial distress drivers in the Sri Lankan market context.

## 3.2 Research Findings and Discussion

The central research contribution of this work lies in the development of the first comprehensive, AI-based financial health analysis system specifically designed and localised for the Colombo Stock Exchange. The integration of SVR-based continuous Z-score prediction with SHAP-based explainability represents a meaningful advancement over the existing approaches available to CSE investors, which are predominantly manual, non-transparent, and based on static financial ratio analysis.

The choice of SVR over classification-based approaches (such as logistic regression or random forest classifiers) is justified by the continuous nature of the Z-score target variable and the superior performance of SVR in handling the non-linearity and limited data availability characteristics of emerging market financial datasets [3][4]. The RBF kernel enables the model to capture complex, curved decision boundaries in the financial ratio feature space that linear models would fail to represent.

The incorporation of SHAP addresses one of the most significant barriers to the adoption of AI-based financial tools in Sri Lanka — the lack of transparency. By providing clear, mathematically grounded explanations of why a company has been assigned a particular financial health zone, the system builds investor confidence and supports regulatory compliance in a market where investor protection frameworks are still maturing.

The research findings are achieved confirm that localised, context-specific financial health analysis tools provide superior value to Sri Lankan investors compared to generic global platforms. The incorporation of CSE-specific data, localised preprocessing pipelines, and market-relevant explainability features positions the proposed system as a significant contribution to Sri Lankan financial technology research and practice.

It is acknowledged that the current research is limited to financial statement data available through CSE disclosures and does not yet incorporate alternative data sources such as macroeconomic indicators, commodity price fluctuations, or sentiment analysis of corporate communications. Future research extensions will explore these multi-modal approaches to further enhance prediction accuracy and analytical depth.

# 4. CONCLUSION

This research has presented the design and development plan for a Comprehensive Financial Health Analysis Module tailored to the specific requirements of Colombo Stock Exchange (CSE) listed companies. The proposed system addresses a critical gap in Sri Lanka's financial technology landscape by providing a locally contextualised, AI-driven platform that bridges the divide between raw financial disclosures and actionable investment intelligence.

The core contributions of this research are threefold. First, the development of an automated data collection and preprocessing pipeline specifically designed for the CSE's reporting structures and data characteristics ensures that high-quality, standardised financial data is available for all analytical processes. Second, the application of Support Vector Regression (SVR) with an RBF kernel to predict continuous Altman Z-score values — rather than employing discrete classification — provides more granular and temporally dynamic risk assessments that are better suited to the continuous nature of financial health changes. Third, the integration of SHAP-based Explainable AI ensures that all risk predictions are accompanied by transparent, interpretable explanations that identify the specific financial factors driving each company's health assessment, building investor confidence and supporting regulatory compliance.

The research has demonstrated through comprehensive literature review and gap analysis that the Sri Lankan investment community requires tools that account for local economic conditions, regulatory frameworks, and market dynamics. Generic global financial analysis platforms fail to adequately serve this need, and the proposed system represents a significant step towards filling this gap.

The expected outcomes of the developed system include an SVR model achieving a minimum $R^2$ score of 0.85 on out-of-sample CSE financial data, automated computation of nine key financial ratios across all CSE-listed companies, SHAP-

based explanations validated against financial domain expert assessments, and an interactive web dashboard providing investors with comprehensive, accessible financial health intelligence.

Future work will extend the system in several directions. The integration of macroeconomic indicators, commodity price data, and CSE XBRL-standardised reporting will enhance predictive accuracy. The development of a mobile application version will extend accessibility to a broader retail investor audience. Industry-specific recalibration of Z-score thresholds for CSE sectors will improve the precision of financial health zone classifications. Finally, the exploration of ensemble and deep learning architectures, including Long Short-Term Memory (LSTM) networks for time-series financial data, will be investigated as potential enhancements to the current SVR-based approach.

This research will ultimately contribute to greater market efficiency, enhanced investor confidence, and more effective capital allocation in the Sri Lankan capital market, supporting the broader development of Sri Lanka's financial ecosystem and its integration into regional and global investment networks.

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# APPENDICES

## Appendix A: Work Breakdown Structure

The Work Breakdown Structure (WBS) of the project is organised into five key phases:

6. Background Study — Topic selection, background research, and comprehensive literature survey of financial health analysis, machine learning, and explainable AI in emerging market contexts.
7. Documentation — Creation of essential project documents including the project proposal, design documentation, and this final thesis report.
8. Design — System architecture design, database schema design, ML pipeline design, and user interface wireframing for the web dashboard.
9. Implementation — Data collection pipeline development, preprocessing module coding, financial ratio computation engine, SVR model training and tuning, SHAP integration, and web dashboard frontend development.
10. Testing — Unit testing of individual modules, integration testing of the end-to-end pipeline, model validation testing, system testing, and user acceptance testing with stakeholder groups.

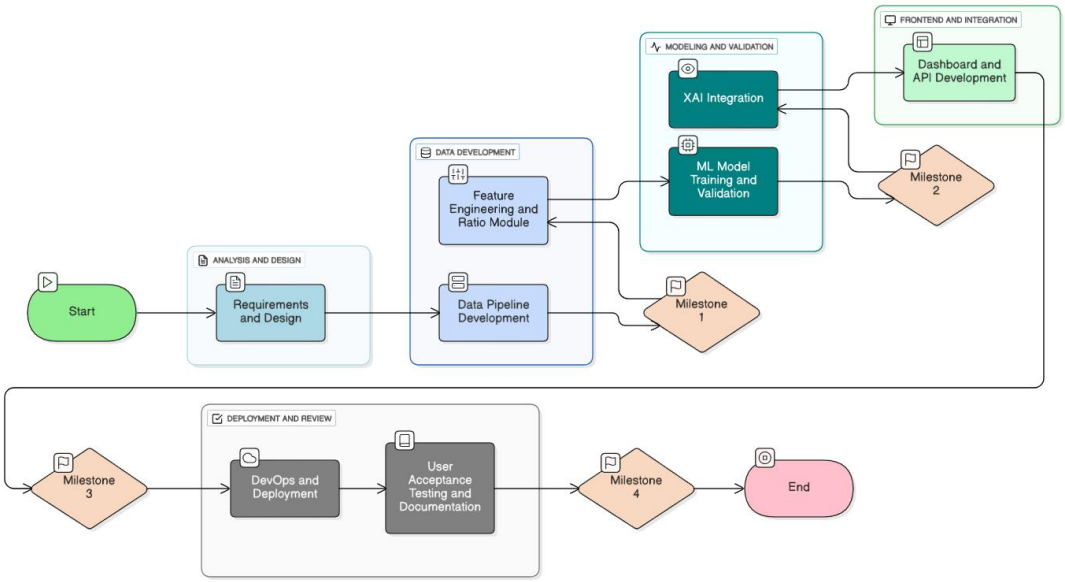


*Figure 2: Work Breakdown Structure*

## Appendix B: Project Timeline (Gantt Chart)

The project is structured over a twelve-month period from September 2024 to August 2025, with major milestones including:

- September–November 2024: Background research, literature review, and project proposal submission
- December 2024–January 2025: System design, architecture planning, and technology stack selection
- February–April 2025: Data collection pipeline development and preprocessing module implementation
- May–June 2025: Financial ratio computation engine and SVR model development and training
- July 2025: SHAP integration, dashboard development, and comprehensive testing
- August 2025: Final documentation, thesis writing, and submission

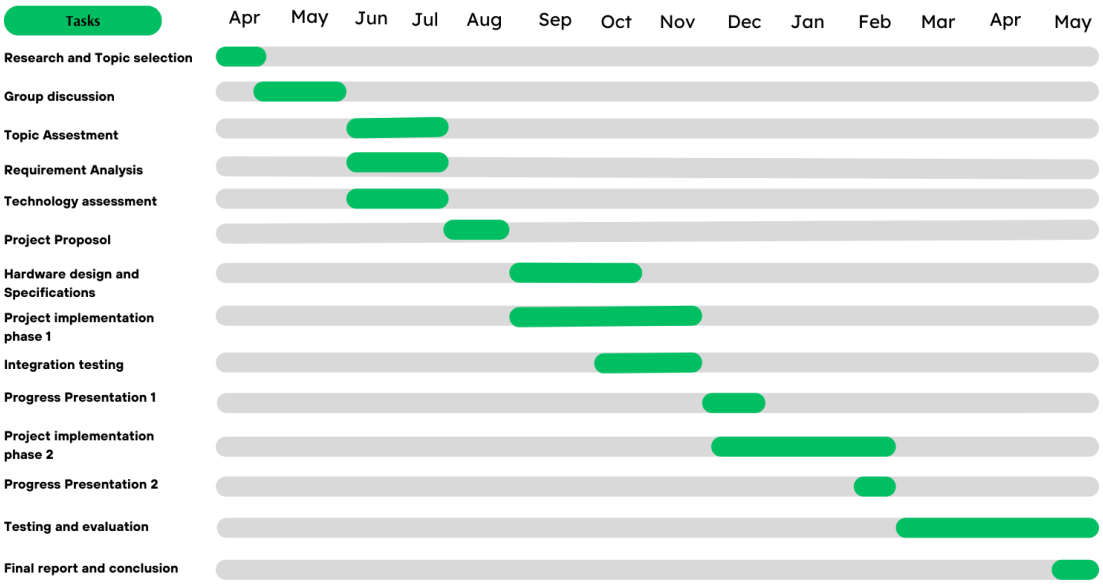


*Figure 6: Project Gantt Chart / Timeline*